4th only

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3) $\sqrt[3]{\sqrt{27}} = \sqrt[3]{\sqrt[2]{27}} = \sqrt[6]{27} = \sqrt[6]{3^3} =$

$= 3^{3/6} = 3^{1/2} = \boxed{\sqrt{3}}$

Must reduce index.

F12 4th CAK

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F12 4th CAR

Lecture 3: Section A.3
Polynomials and Factoring

Polynomials

coef.

<u>Def.</u> A **polynomial in** x is an expression of the form

$$-\frac{2}{3}x^5 + \sqrt{3}\,x - 4x^0$$

$$a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

where $a_n, \cdots, a_0$ are real numbers and n is a nonnegative integer. exponents: 0, 1, 2, 3, ...

NOTE:

1) $a_n, \cdots, a_0$ are the **coefficients** of each term. If $a_n \neq 0$, the polynomial has **degree** n, and a_n is called the **leading coefficient**. $-\frac{2}{3}$ 5

2) Polynomials with one, two, and three terms are called **monomials, binomials,** and **trinomials,** respectively.

3) In **standard form**, a polynomial is written with descending powers of x.

4) A polynomial with all $a_i = 0$ $(i = 0, 1, \cdots .n)$ is called the **zero polynomial**. No degree is assigned to the zero polynomial. $0x^2 = 0$

exponents: 0, 1, 2, 3, ..

ex. State whether each is a polynomial. For each polynomial, rewrite it in standard form and find its degree.

yes 1) $-2 = -2x^0$ degree 0

no 2) $2x^4 - 1 + \frac{3}{x} = 3x^{-1}$

no 3) $2x^5 + 4x - \sqrt{2x} = \sqrt{2}\sqrt{x} = \sqrt{2}\,x^{1/2}$

yes 4) $5x - 7 + \sqrt{3}x^8 - 4x^5$ degree 8

$\sqrt{3}x^8 - 4x^5 + 5x - 7$

Operations with Polynomials

1. Addition and Subtraction of Polynomials:

Combine like terms (same variable to the same power) by adding their coefficients.

ex. Find the difference:

$$(x^3 + 6x^2 - 4x + 7) - (3x^2 + 2x - 4) =$$

$x^3 + 6x^2 - 4x + 7 - 3x^2 - 2x + 4 =$

$\boxed{x^3 + 3x^2 - 6x + 11}$

2. Multiplication of Polynomials:

Use the Properties of Exponents, Commutative, Associative and Distributive Properties to find the product of two polynomials.

ex. $(-3x^8)(6x^5) = -3 \cdot x^8 \cdot 6 \cdot x^5 = -18x^{13}$

ex. Find the product: $(3x-4)(x^3-2x-6)$

$x^3 - 2x - 6$

$3x - 4$

$3x^4 - 6x^2 - 18x - 4x^3 + 8x + 24 =$

$3x^4 - 4x^3 - 6x^2 - 10x + 24$

First, outer, inner, last

When multiplying two binomials we use **FOIL**.

ex. $(2x-5)(4x+7) = 8x^2 + 14x - 20x - 35 =$

$8x^2 - 6x - 35$

Special Products Short-cuts

* 1. Difference of Two Squares

FOIL →

$$(u+v)(u-v) = u^2 - v^2$$

* 2. Square of a Binomial

FOIL

$$(u+v)^2 = (u+v)(u+v) = u^2 + 2uv + v^2$$

$$(u-v)^2 = (u-v)(u-v) = u^2 - 2uv + v^2$$

NOTE: $(u+v)^2 \neq u^2 + v^2$

ex. Find $(3x+\sqrt{5})^2 = (3x)^2 + 2(3x)(\sqrt{5}) + (\sqrt{5})^2$

$$= \boxed{9x^2 + 6\sqrt{5}x + 5}$$

ex. Find $((x^2+x)+2)((x^2+x)-2) =$

Let $u = x^2+x$

$v = 2$

$$(u+v)(u-v) = u^2 - v^2$$

$$= (x^2+x)^2 - (2)^2$$

$$= (x^2)^2 + 2x^2(x) + x^2 - 4$$

$$= \boxed{x^4 + 2x^3 + x^2 - 4}$$

If not memorized

$(u+v)^3 = (u+v)(u+v)(u+v) =$

* 3. <u>Cube of a Binomial</u>

or $(x^2 + (-2))^3$

$(u+v)^3 = u^3 + 3u^2v + 3uv^2 + v^3$

$(u-v)^3 = u^3 - 3u^2v + 3uv^2 - v^3$

<u>ex.</u> Find $(x^2-2)^3 = (x^2)^3 - 3(x^2)^2(2) + 3(x^2)(2)^2 - 2^3$

$u = x^2$

$v = 2$

$= x^6 - 6x^4 + 12x^2 - 8$

Factoring

$$\Longrightarrow$$

$$x^2 - 4 = (x-2)(x+2)$$

$$\Longleftarrow$$

To **factor** a polynomial is to express it as the **product** of polynomial expressions, called **factors**.

A polynomial which can not be written as the product of factors other than itself and 1 is called **prime**. A polynomial is **factored completely** if it is written as the product of prime polynomials only.

Factoring out the Greatest Common Factor (GCF)

Use the distributive property

$$ab + ac = a(b + c)$$

to factor out monomial factors common to each term.

<u>**ex.**</u> Factor completely:

1) $3x^2 - 6x = \boxed{3x(x-2)}$

2) $8x^4y^2 + 6x^3y^3 - 2xy^4 = \boxed{2xy^2(4x^3 + 3x^2y - y^2)}$

$\cancel{[(x-2)-2] \neq (x-2)(-2)}$

3) $5(x-2)^4 - 10(x-2)^3 = 5(x-2)^3[(x-2) - 2]$

$= \boxed{5(x-2)^3(x-4)}$

4) $y + 6$

prime

Factoring Special Polynomial Forms

1. Difference of Two Squares

$$u^2 - v^2 = (u+v)(u-v)$$

2. Perfect Square Trinomial

$$u^2 + 2uv + v^2 = (u+v)^2$$

$$u^2 - 2uv + v^2 = (u-v)^2$$

ex. Factor completely:

$u = x+2$
$v = y$

1) $(x+2)^2 - y^2 = \boxed{(x+2+y)(x+2-y)}$

$u^2 - v^2 = (u+v)(u-v)$

$(2x)^2 - (3)^2$

$u = 4x^2$
$v = 9$

2) $16x^4 - 81 = (4x^2)^2 - (9)^2 = (4x^2+9)(4x^2-9)$

$= \boxed{(4x^2+9)(2x+3)(2x-3)}$

$x \quad 4$

3) $x^2 - 8x + 16$

$= (x-4)^2$ check: $(x-4)(x-4)$ FOIL

$3y \quad 5$

4) $9y^2 + 30y + 25 = (3y+5)^2$

check: $(3y+5)(3y+5)$ FOIL

3. Sum and Difference of Cubes

$$u^3 + v^3 = (u+v)(u^2 - uv + v^2)$$

$$u^3 - v^3 = (u-v)(u^2 + uv + v^2)$$

ex. Factor completely:

1) $27x^3 - 1 = (3x)^3 - (1)^3 =$

$= (3x-1)((3x)^2 + 3x(1) + 1^2) =$

$\boxed{(3x-1)(9x^2+3x+1)}$

2) $y^6 + 8$

$(y^2)^3 + (2)^3 = \boxed{(y^2+2)(y^4 - 2y^2 + 4)}$

Factoring Trinomial $x^2 + Bx + C$

$A = 1$

$$x^2 + Bx + C = (x + a)(x + b)$$

where $ab = C$

and $a + b = B$

$3, -10 \rightarrow$ x to give -30

$\rightarrow$ add to give -7

ex. Factor: $x^2 - 7x - 30 =$

$\boxed{(x+3)(x-10)}$

Can FOIL to check.

not 1

Can do by trial + error.

Factoring Trinomial $Ax^2 + Bx + C,\ A \neq 1$

<u>**ex.**</u> Factor: 1) $2x^2 - 5x + 2 = (2x-2)(x-1)$

$\boxed{(2x-1)(x-2)}$

2) $6x^2 - x - 12$

HW

$\boxed{(3x+4)(2x-3)}$

Factoring by Grouping

We use this method for polynomials with more than 3 terms. Group terms with common factors, and then factor the result.

<u>**ex.**</u> Factor: $xy - 2y - 4x + 8 = (xy - 2y) + (-4x + 8)$

Let $w = x - 2$

$= y(x-2) - 4(x-2)$

$yw - 4w =$

$w(y-4)$ now sub back

$(x-2)(y-4)$

$= \boxed{(x-2)(y-4)}$